

Assignment 4 Solutions

4.1 $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \sim \begin{bmatrix} \boxed{1} & -1 \\ 0 & \boxed{1} \end{bmatrix}$ Linearly independent, 1 pivot per column
(also two nonzero vectors are L.D. iff they are colinear)

4.2 $\begin{bmatrix} 1 & -1 & -3 \\ -1 & 4 & 9 \\ 2 & -3 & -8 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -3 \\ 0 & 3 & 6 \\ 0 & -1 & -2 \end{bmatrix} \sim \begin{bmatrix} \boxed{1} & -1 & -3 \\ 0 & \boxed{1} & 2 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$ $x_1 = x_3$
 $x_2 = -2x_3$
 x_3 is free
 $\vec{v}_1 - 2\vec{v}_2 + \vec{v}_3 = \vec{0}$

4.7 $\begin{bmatrix} 2 & -3 & 3 & 8 \\ 0 & -2 & -2 & 7 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & -3 & 3 & 0 \\ 0 & -2 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & -3 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & 0 & 6 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$$x_1 = -3x_3$$

$$x_2 = -x_3$$

$$x_3 \text{ is free}$$

$$x_4 = 0$$

$$3\vec{v}_1 + \vec{v}_2 - \vec{v}_3 = \vec{0}$$

5.10 $T(-3\vec{v}_1 - \vec{v}_2 - 2\vec{v}_3) = -3T(\vec{v}_1) - T(\vec{v}_2) - 2T(\vec{v}_3) =$

$$-3 \begin{bmatrix} -6 \\ 3 \\ -2 \\ -10 \end{bmatrix} - \begin{bmatrix} -5 \\ 1 \\ -2 \\ 9 \end{bmatrix} - 2 \begin{bmatrix} 8 \\ -7 \\ 6 \\ 6 \end{bmatrix} = \begin{bmatrix} 18 + 5 - 16 \\ -9 - 1 + 14 \\ 6 + 2 - 12 \\ 30 - 9 - 12 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -4 \\ 9 \end{bmatrix}$$

5.12 $\begin{bmatrix} 3 & -3 & 21 \\ 2 & 1 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 7 \\ 2 & 1 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 7 \\ 0 & 3 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 7 \\ 0 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & -2 \end{bmatrix}$ $T\left(\begin{bmatrix} 5 \\ -2 \end{bmatrix}\right) = \begin{bmatrix} 21 \\ 8 \end{bmatrix}$
unique solution

$$5.13 \quad \begin{bmatrix} 1 & -1 & 2 & 6 \\ 2 & -1 & 3 & 12 \\ -2 & 2 & -3 & -10 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 2 & 6 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{bmatrix} \quad T \begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 12 \\ -10 \end{pmatrix}$$

unique solution

$$5.15 \quad \begin{matrix} \vec{v}_1 & \vec{v}_2 & \vec{v}_3 & \vec{u} \\ \begin{bmatrix} -3 & -9 & 9 & 30 \\ 9 & -5 & -1 & 0 \\ 5 & 0 & -2 & -7 \\ -4 & -7 & 4 & 12 \end{bmatrix} \end{matrix} \sim \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \vec{u} = -\vec{v}_1 - 2\vec{v}_2 + \vec{v}_3$$

$$T(\vec{u}) = T(-\vec{v}_1 - 2\vec{v}_2 + \vec{v}_3) = -T(\vec{v}_1) - 2T(\vec{v}_2) + T(\vec{v}_3) = - \begin{bmatrix} -4 \\ 3 \\ 4 \end{bmatrix} - 2 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ -3 \\ 7 \end{bmatrix}$$

$$= \begin{bmatrix} 4 + 3 \\ -3 - 2 - 3 \\ -4 - 4 + 7 \end{bmatrix} = \begin{bmatrix} 7 \\ -8 \\ -1 \end{bmatrix}$$

4.8 False, $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

4.9 True, pivot per column, all entries other than the pivot in a pivot column must be 0.

4.10 False, $v_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $v_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $v_3 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

4.11 False, $\begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$

4.12 True, if $a\vec{v}_1 + b\vec{v}_2 = \vec{0}$ is a nontrivial dependence relation then so is $a\vec{v}_1 + b\vec{v}_2 + 0\vec{v}_3 = \vec{0}$

5.40 False, $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \notin \text{ran}(T)$ where T is $\vec{x} \mapsto A\vec{x}$

5.41 False, $A \vec{v}$, need A to be $n \times m$.
 $(n \times m) \mathbb{R}^m$

4.22 (a) $\vec{v}_4 \in \text{span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ because $[\vec{v}_1 \ \vec{v}_2 \ \vec{v}_3 \ \vec{v}_4]$ thought of as an augmented matrix represents a consistent system.

$$\begin{bmatrix} 1 & 1 & 3 & -2 \\ 0 & 4 & -4 & 12 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 & -2 \\ 0 & 4 & -4 & 12 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & -5 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -9 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$-9\vec{v}_1 + 4\vec{v}_2 + \vec{v}_3 = \vec{v}_4$$

(b) $\begin{bmatrix} 1 & 3 & -2 \\ 4 & -4 & 12 \\ 0 & 3 & 3 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -9 \\ 1 & 0 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & -9 \\ 0 & 0 & 0 \end{bmatrix}$ 1 pivot per column
 Linearly independent

4.23 $\begin{bmatrix} 1 & 1 & -2 \\ 9 & 4 & h \\ -3 & 3 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & -2 \\ 1 & -1 & 2 \\ 9 & 4 & h \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & -2 \\ 0 & 1 & -2 \\ 0 & -5 & h+8 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & h+8 \end{bmatrix}$ 1 pivot / column
 $\Downarrow$
 $h = -8$

4.24 $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ $\vec{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ $\vec{v}_3 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$ $\vec{v}_1 \notin \text{span}\{\vec{v}_2, \vec{v}_3\}$
 but $0\vec{v}_1 + \vec{v}_2 + \vec{v}_3 = \vec{0}$

5.44 (a)

$$\begin{bmatrix} 1/8 \\ 1/8 \\ 1/8 \\ 1/8 \\ 1/5 \\ 1/10 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \\ 0 \\ 0 \end{bmatrix}$$

5.53 (a)

$$\begin{aligned} T \left(c \begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) &= T \begin{bmatrix} cx \\ cy \\ cz \end{bmatrix} = \begin{bmatrix} ((cx)^3 + (cy)^3 + (cz)^3)^{1/3} \\ cy + cz \end{bmatrix} = \begin{bmatrix} (c^3 x^3 + c^3 y^3 + c^3 z^3)^{1/3} \\ c(y+z) \end{bmatrix} \\ &= \begin{bmatrix} (c^3 (x^3 + y^3 + z^3))^{1/3} \\ c(y+z) \end{bmatrix} = \begin{bmatrix} c(x^3 + y^3 + z^3)^{1/3} \\ c(y+z) \end{bmatrix} \\ &= c \begin{bmatrix} x^3 + y^3 + z^3 \\ y+z \end{bmatrix} = c T \begin{bmatrix} x \\ y \\ z \end{bmatrix} \end{aligned}$$

(b) Not linear.

$$\begin{aligned} T \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) &= T \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2^{1/3} \\ 1 \end{bmatrix} \\ T \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad T \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ T \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + T \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} &= \begin{bmatrix} 2 \\ 1 \end{bmatrix} \end{aligned}$$

not equal